

None



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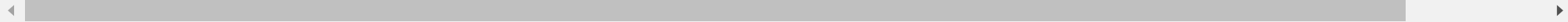
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1st Edition

Why This E-book?

"The aim of this ebook is to give you the 'aha' moment right away at the start of learning a new c

- Practical step By Step Guide With Simple Examples
- Visual Illustrations and Interactive
- Simple Datasets
- Comprehensive Coverage(pandas Documentation used as reference)

Introduction

pandas.crosstab is a powerful function in the Pandas library used for creating contingency tables, which display the frequency distribution of variables. It allows you to cross-tabulate two or more factors, showing the counts or proportions of each combination of categories. You can easily add margins (totals) and normalize the data to show percentages or fractions, making it a versatile tool for data analysis and summarization.

The "pivot_table" syntax

```
pandas.crosstab(index, columns, values=None, rownames=None,
colnames=None, aggfunc=None, margins=False, margins_name='All',
dropna=True, normalize=False)
```

The index parameter

Specifies values to group by in the rows.

This parameter specifies the column(s) whose unique values will be used to form the rows of the resulting cross-tabulation.

The "index" parameter accepts array-like, Series, or list of arrays/Series

```
import numpy as np
import pandas as pd
```

Count of smokers and non-smokers for each sex

DataFrame

	sex	smoker	time
0	Female	Yes	Dinner
1	Male	No	Dinner
2	Female	Yes	Lunch
3	Male	No	Lunch
4	Female	Yes	Dinner
5	Male	No	Dinner

Summary:

	sex	smoker	time	Dinner	Lunch
Female		No		0	1
		Yes		3	1

Code snippet:

```
pd.crosstab(
    index=[df['sex'], df['smoker']],
```

	sex	smoker	time
6	Female	No	Lunch
7	Male	Yes	Dinner
8	Female	Yes	Dinner
9	Male	No	Dinner

	sex	smoker	time	Dinner	Lunch
Male	No		3	1	
	Yes		1	0	

```
columns=df['time']
)
```

Code Snippet:

```
data = {
    'sex': ['Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male'],
    'smoker': ['Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No'],
    'time': ['Dinner', 'Dinner', 'Lunch', 'Lunch', 'Dinner', 'Dinner', 'Lunch', 'Dinner', 'Dinner', 'Dinner']
}
df = pd.DataFrame(data)
df
```



df:

	sex	smoker	time
0	Female	Yes	Dinner
1	Male	No	Dinner
2	Female	Yes	Lunch
3	Male	No	Lunch
4	Female	Yes	Dinner
5	Male	No	Dinner
6	Female	No	Lunch
7	Male	Yes	Dinner
8	Female	Yes	Dinner
9	Male	No	Dinner

The **columns** parameter

Specifies Values to group by in the columns.

The "columns" parameter accepts array-like, Series, or list of arrays/Series

Counts of combinations of smoker and time for each sex category

DataFrame

	sex	smoker	time
0	Female	Yes	Dinner
1	Male	No	Dinner
2	Female	Yes	Lunch
3	Male	No	Lunch
4	Female	Yes	Dinner
5	Male	No	Dinner
6	Female	No	Lunch
7	Male	Yes	Dinner
8	Female	Yes	Dinner
9	Male	No	Dinner



Summary:

	smoker		No		Yes	
	time		Dinner	Lunch	Dinner	Lunch
sex						
Female		0	1	3	1	
Male		3	1	1	0	

Code snippet:

```
pd.crosstab(
    index=df['sex'],
    columns=[df['smoker'], df['time']]
)
```

df:

Code Snippet:

```
data = {
    'sex': ['Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male'],
    'smoker': ['Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No'],
    'time': ['Dinner', 'Dinner', 'Lunch', 'Lunch', 'Dinner', 'Dinner', 'Lunch', 'Dinner', 'Dinner', 'Dinner']
}
df = pd.DataFrame(data)
df
```



	sex	smoker	time
0	Female	Yes	Dinner
1	Male	No	Dinner
2	Female	Yes	Lunch
3	Male	No	Lunch
4	Female	Yes	Dinner
5	Male	No	Dinner
6	Female	No	Lunch
7	Male	Yes	Dinner
8	Female	Yes	Dinner
9	Male	No	Dinner

The values parameter

Array of values to aggregate according to the factors. Requires aggfunc be specified.
The "values" parameter accepts array-like and is optional.

Maximum tip amount for each combination of 'sex' and 'smoker' status

DataFrame

Code snippet:

	sex	smoker	tip
0	Female	Yes	5.5
1	Female	Yes	3.0
2	Male	No	4.5
3	Male	No	3.5
4	Female	No	4.0
5	Female	No	5.0
6	Male	Yes	6.0
7	Male	Yes	2.5

**Summary:**

	smoker	No	Yes
sex			
Female		5.0	5.5
Male		4.5	6.0

```
pd.crosstab(  
    index=df['sex'],  
    columns=df['smoker'],  
    values=df['tip'],  
    aggfunc='max'  
)
```

Code Snippet:

```
data = {
    'sex': ['Female', 'Female', 'Male', 'Male', 'Female', 'Female',
            'Male', 'Male'],
    'smoker': ['Yes', 'Yes', 'No', 'No', 'No', 'No', 'Yes', 'Yes'],
    'tip': [5.5, 3.0, 4.5, 3.5, 4.0, 5.0, 6.0, 2.5]
}
df = pd.DataFrame(data)
df
```

df:

	sex	smoker	tip
0	Female	Yes	5.5
1	Female	Yes	3.0
2	Male	No	4.5
3	Male	No	3.5
4	Female	No	4.0
5	Female	No	5.0
6	Male	Yes	6.0
7	Male	Yes	2.5

The rownames parameter

"rownames" is used to label the rows in the resulting table.

The "rownames " parameter accepts sequence and it is None by default.

If passed, must match number of row arrays passed.

Rename the row index to 'Sexoo'**DataFrame**

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No
7	Male	Yes
8	Female	Yes
9	Male	No

**Summary:**

smoker	No	Yes
Sexoo		
Female	1	4
Male	3	2

Code snippet:

```
pd.crosstab(
    index=df['sex'],
    columns=[df['smoker']],
    rownames = ["Sexoo"]
)
```

Code Snippet:

```
data = {
    'sex': ['Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male'],
    'smoker': ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No']
}
```

**df:**

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No

```
df = pd.DataFrame(data)
```

```
df
```

	sex	smoker
7	Male	Yes
8	Female	Yes
9	Male	No

The `colnames` parameter

"colnames" is used to label the cols in the resulting table.

The "colnames " parameter accepts sequence and it is None by default.

If passed, must match number of col arrays passed.

Rename the column index(smoker) to 'Fumador'

DataFrame

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No



Summary:

	Fumador	No	Yes
sex			
Female		1	4
Male		3	2

Code snippet:

```
pd.crosstab(
    index=df['sex'],
    columns=[df['smoker']],
```

	sex	smoker
7	Male	Yes
8	Female	Yes
9	Male	No

```
colnames = ["Fumador"]  
)
```

Code Snippet:

```
data = {  
    'sex': ['Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male'],  
    'smoker': ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'Yes', 'No']  
}  
df = pd.DataFrame(data)  
df
```

df:

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No
7	Male	Yes
8	Female	Yes
9	Male	No

The **aggfunc** parameter

Accepts a function and it is optional..

If specified, requires values be specified as well.

Sum and the mean of tips for each combination of 'sex' and 'smoker'

DataFrame

	sex	smoker	tip
0	Female	Yes	5.5
1	Female	Yes	3.0
2	Male	No	4.5
3	Male	No	3.5
4	Female	No	4.0
5	Female	No	5.0
6	Male	Yes	6.0
7	Male	Yes	2.5



Summary:

	sum		mean	
smoker	No	Yes	No	Yes
sex				
Female	9.0	8.5	4.5	4.25
Male	8.0	8.5	4.0	4.25

Code snippet:

```
pd.crosstab(
    index=df['sex'],
    columns=df['smoker'],
    values=df['tip'],
    aggfunc=['sum', 'mean']
)
```

The **margins** parameter

The "margins" parameter Adds row/column margins (subtotals).
It is boolean and defaults to False.

Sum and the mean of tips for each combination of 'sex' and 'smoker', and also include row and column totals

DataFrame

	sex	smoker	tip
0	Female	Yes	5.5
1	Female	Yes	3.0
2	Male	No	4.5
3	Male	No	3.5
4	Female	No	4.0
5	Female	No	5.0
6	Male	Yes	6.0
7	Male	Yes	2.5



Summary:

	sum			mean		
smoker	No	Yes	All	No	Yes	All
sex						
Female	9.0	8.5	17.5	4.50	4.25	4.375
Male	8.0	8.5	16.5	4.00	4.25	4.125
All	17.0	17.0	34.0	4.25	4.25	4.250

Code snippet:

```
pd.crosstab(
    index=df['sex'],
    columns=df['smoker'],
    values=df['tip'],
    aggfunc=['sum', 'mean'],
    margins = True
)
```

The **margins_name** parameter

The "margins_name" parameter specifies the name of the row or column that will contain the totals when margins=True."

It is string and By default, the name is set to 'All'.

Sum and the mean of tips for each combination of 'sex' and 'smoker', and also include row and column totals

Code snippet:

DataFrame

	sex	smoker	tip
0	Female	Yes	5.5
1	Female	Yes	3.0
2	Male	No	4.5
3	Male	No	3.5
4	Female	No	4.0
5	Female	No	5.0
6	Male	Yes	6.0
7	Male	Yes	2.5

Summary:

		sum			mean	
smoker	No	Yes	Subtotal	No	Yes	Subtotal
sex						
Female	9.0	8.5	17.5	4.50	4.25	4.375
Male	8.0	8.5	16.5	4.00	4.25	4.125
Subtotal	17.0	17.0	34.0	4.25	4.25	4.250

```
pd.crosstab(
    index=df['sex'],
    columns=df['smoker'],
    values=df['tip'],
    aggfunc=['sum', 'mean'],
    margins = True,
    margins_name = 'Subtotal'
)
```

The **dropna** parameter

The "dropna" parameter does not include columns whose entries are all NaN. If True, rows with a NaN value in any column will be omitted before computing margins. It is a boolean and defaults to True.

Show the total sales (sum of 'Price') for each manager across different products.

df:

	Manager	Product	Quantity	Price	Status
0	Debra	CPU	2.0	600.0	None

When dropna=False:

	Product	CPU	RAM	nan	All
Manager					
Debra	600.0	100.0	NaN	700.0	
Fred	300.0	0.0	NaN	300.0	
NaN	NaN	NaN	0.0	NaN	

When dropna=True (default):

	Product	CPU	RAM	All
Manager				
Debra	600.0	100.0	700.0	
Fred	300.0	0.0	300.0	
All	900.0	100.0	1000.0	

	Manager	Product	Quantity	Price	Status
1	Debra	RAM	NaN	100.0	None
2	Fred	CPU	1.0	300.0	None
3	Fred	RAM	3.0	NaN	None
4	None	None	NaN	NaN	None



	Product	CPU	RAM	nan	All
Manager					
All	900.0	100.0	NaN	1000.0	

```
pd.crosstab(
    index=df['Manager'],
    columns=df['Product'],
    values=df['Price'],
    aggfunc='sum',
    margins=True,
    dropna = False
)
```

```
pd.crosstab(
    index=df['Manager'],
    columns=df['Product'],
    values=df['Price'],
    aggfunc='sum',
    margins=True,
    dropna =True
)
```

The **normalize** parameter

The "normalize" parameter is boolean.

It accepts {'all', 'index', 'columns'}, or {0,1} and defaults to False.

Its purpose is to normalize(Proportion) by dividing all values by the sum of values.

Code Snippet:

```
data = {
    'sex': ['Female', 'Male', 'Female', 'Male', 'Female', 'Male',
            'Female', 'Male', 'Female', 'Male'],
    'smoker': ['Yes', 'No', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'No', 'Yes']
}
```

df:

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No

```
es', 'Yes', 'No'],
    'time': ['Dinner', 'Dinner', 'Lunch', 'Lunch', 'Dinner',
'Dinner', 'Lunch', 'Dinner', 'Dinner', 'Dinner']
}
df = pd.DataFrame(data)
df
```



	sex	smoker
4	Female	Yes
5	Male	No
6	Female	No
7	Male	Yes
8	Female	Yes
9	Male	No

If passed 'all' or True, will normalize over all values.

Generate a table of sex and smoker status in a DataFrame, showing proportions relative to the entire dataset

df:

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No
7	Male	Yes

Without nnormalize:

	smoker	No	Yes	Total
sex				
Female		1	4	5
Male		3	2	5
Total		4	6	10

When normalize = True or "all":

	smoker	No	Yes
sex			
Female		0.1	0.4
Male		0.3	0.2



```
pd.crosstab(
    index=[df['sex']],
    columns=[df['smoker']],
    margins = True,
```

```
pd.crosstab(
    index=[df['sex']],
    columns=[df['smoker']],
```

	sex	smoker
8	Female	Yes
9	Male	No

```
margins_name='Total'
)

normalize = True
)
```

If passed 'index' will normalize over each row.

df:

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No
7	Male	Yes
8	Female	Yes
9	Male	No

Without nnormalize:

	smoker	No	Yes	Total
sex				
Female		1	4	5
Male		3	2	5
Total		4	6	10

When normalize = index or "0":

	smoker	No	Yes
sex			
Female		0.2	0.8
Male		0.6	0.4

```
pd.crosstab(
    index=[df['sex']],
    columns=[df['smoker']],
    margins = True,
    margins_name='Total'
)

pd.crosstab(
    index=[df['sex']],
    columns=[df['smoker']],
    normalize = 'index'
)
```

If passed 'columns' will normalize over each column.

df:

	sex	smoker
0	Female	Yes

Without normalize:

When normalize = 'index' or 1:

	sex	smoker
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No
7	Male	Yes
8	Female	Yes
9	Male	No



	smoker	No	Yes	Total
sex				
Female		1	4	5
Male		3	2	5
Total		4	6	10

```
pd.crosstab(  
    index=[df['sex']],  
    columns=[df['smoker']],  
    margins = True,  
    margins_name='Total'  
)
```

pandas.crosstab

	smoker	No	Yes
sex			
Female		0.25	0.666667
Male		0.75	0.333333

```
pd.crosstab(  
    index=[df['sex']],  
    columns=[df['smoker']],  
    normalize = 'index'  
)
```

If margins is True, will also normalize margin values.

df:

	sex	smoker
0	Female	Yes
1	Male	Yes
2	Female	Yes
3	Male	No
4	Female	Yes
5	Male	No
6	Female	No
7	Male	Yes
8	Female	Yes
9	Male	No



Without nnormalize:

	smoker	No	Yes	Total
sex				
Female		1	4	5
Male		3	2	5
Total		4	6	10

```
pd.crosstab(  
    index=[df['sex']],  
    columns=[df['smoker']],  
    margins = True,  
    margins_name='Total'  
)
```

When normalize = True or "all":

	smoker	No	Yes	Total
sex				
Female		0.1	0.4	0.5
Male		0.3	0.2	0.5
Total		0.4	0.6	1.0

```
pd.crosstab(  
    index=[df['sex']],  
    columns=[df['smoker']],  
    normalize = True,  
    margins = True,  
)
```

```
margins_name='Total'  
)
```

Project(Real Life application)

[pandas.crosstab](https://interactivechaos.com/en/python/function/pandascrosstab) (<https://interactivechaos.com/en/python/function/pandascrosstab>)

Sources & References

[pandas.crosstab Documentation](https://pandas.pydata.org/docs/reference/api/pandas.crosstab.html#pandas.crosstab)

(<https://pandas.pydata.org/docs/reference/api/pandas.crosstab.html#pandas.crosstab>)

[pandas.crosstab](https://interactivechaos.com/en/python/function/pandascrosstab) (<https://interactivechaos.com/en/python/function/pandascrosstab>)

Author Biography

Kichere Magubu is a data enthusiast and content creator.

Other E-books

Download (https://github.com/kicherethedatascientist/My_e-Books)

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Please take a minute to write me an honest review. Your support means the world to me!

Thank you!

```
In [ ]: #print("The cell to convert jupyter notebook to html")
        #!jupyter nbconvert --to hide_code_html "pandas.crosstab.ipynb"
```